Object-Centric Grounding for End-to-End Driving

Master's thesis progress and research stay proposal

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Agenda

1. **Master's thesis**
   - Research problem and objectives
   - Technical approach and methodology
   - Progress, preliminary results and how I read them
2. **Research stay proposal**
   - Motivation and the research gap
   - Hypotheses, methodology, expected contributions
3. **Connection between the two**
   - What the thesis delivers that the proposal needs
   - Scientific questions for the stay, fit with the lab

Outline

1 Master's Thesis

2 Research Stay Proposal

3 Connection

Trajectory prediction

Modular stacks detect, then track, then predict[1, 2].

- The interface between stages is just a list of boxes and identities
- Whatever information not encoded in a box is discarded
- *Errors propagate forward* and cannot be corrected downstream

End-to-end perception and prediction removes that interface.
ViP3D[3] carries persistent agent queries through time, and the trajectory decoder reads the query features.

Problem with ViP3D's Agent Queries

- learned embedding of fixed size
- identical in every frame, scene
- reference points come from that embedding (so Query X starts in the same place on an empty motorway and at a busy intersection)

Objectives

Goal of the thesis

A deep learning architecture for end-to-end perception and prediction in which the agent queries originate in the point cloud rather than in a learned embedding.

3 research questions:

1. What does the query-based formulation achieve on the queries **derived from LiDAR**?
2. Once the queries already carry geometric evidence, what **does the camera still contribute**, and to which metrics and classes?
3. A query that starts near its object **needs less layers to correct**. What does that buy in decoder depth and in inference latency?

Query initialisation is interesting

Input-dependent reference points are known to be better and to reduce the decoder depth needed for convergence.

Our reading of the detection literature:

- **DETR**[4] queries are learned vectors with no explicit position at all.
- **Deformable DETR**[5] makes the position explicit as a reference point, predicted from the query embeddin - **input-independent**
- Its two-stage variant takes reference points from the centres of the top-scoring encoder proposals instead, and outperforms the single-stage one.
- **TransFusion**[6] do the initialisation in BEV, heatmap peaks into object queries.

Nobody has applied this to end-to-end perception and prediction, where the query must also survive across frames.

So how is the query generated?

A class-wise heatmap is predicted over the BEV grid with CenterPoint-style[7] Gaussian targets. For each of the top-K peaks:

- the reference point is the metric position of the peak,
- the content is the BEV feature sampled at that cell plus an embedding of its class.

What makes this work with tracking

- Only the slots left free by the tracker are filled
- Agents already being followed keep the queries they carry (suppression)
- initialisation concerns newly appearing objects alone.

Local-max NMS is 3×3 and excludes pedestrian and bicycle classes

Methodology

Training, two stages following the TransFusion recipe[6]

- Stage 1: LiDAR model alone, 20 epochs
- Stage 2: stage 1 frozen, camera layer on top, 6 epochs (so fusion cannot destabilise)
- No data augmentation in any run (future work)

Data, nuScenes[8]

- 700 training and 150 validation scenes, keyframes at 2 Hz
- Persistent instance identities, three-frame clips, 6 s futures
- Seven classes: car, truck, bus, trailer, motorcycle, bicycle, pedestrian

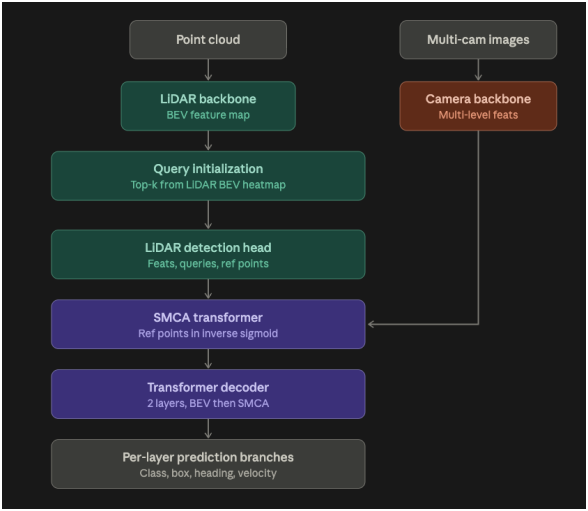
Metrics

- EPA[3] is the primary criterion
- minADE, minFDE, MR cover matched agents only
- Detection and tracking metrics alongside

Protocol

- Identical code, thresholds, class set
- Best epoch by EPA
- Latency: one fixed frame replayed, over 200 timed iterations after warm-up, protocol the published rows use

InitVip3D Architecture



Queries are born from the LiDAR BEV heatmap rather than from a learned embedding.

Two decoder layers refine them, deformable BEV attention and then camera SMCA.

Preliminary results

Table: nuScenes validation, seven classes, 6 s horizon, $K = 6$ modes. The top block is published numbers, the lower two are my stage-1 LiDAR-only and stage-2 fusion runs, every one of mine at effective batch 8 under identical evaluation code.

Run	Queries	EPA ↑	minADE ↓	minFDE ↓	MR ↓	Params (M)	Latency (ms) ↓
ViP3D *as published*[3] (baseline)	300	0.245	1.47	2.21	0.237	41.77	64.3
Li-ViP3D++ *as published*[9]	300	0.505	1.11	1.80	0.209	—	76.28
Stage 1, LiDAR only	300	**0.293**	0.956	1.776	0.183	40.80	90.0
Stage 1, LiDAR only	100	0.256	1.022	1.937	0.239	40.80	81.1
Stage 1, LiDAR only, 5 sweeps	300	0.263	1.001	1.947	0.204	40.80	58.2
Stage 2, + SMCA camera	300	**0.318**	**0.882**	**1.590**	**0.176**	41.66	130.3

Preliminary results: detection

Table: Per-class detection AP, same conditions as above. stage-1 LiDAR-only at effective batch 8, epoch 14. The three published columns are Table III of Li-ViP3D++[9] under its original-input setting.

AP ↑	ViP3D[3]	Li-ViP3D[10]	Li-ViP3D++[9]	**Stage 1 LiDAR only**
car	0.590	0.590	**0.717**	0.433
pedestrian	0.311	0.305	**0.433**	0.381
truck	0.324	0.342	**0.515**	0.095
bus	0.342	0.335	**0.619**	0.000
motorcycle	0.268	0.278	**0.355**	0.000
trailer	0.104	0.114	**0.227**	0.000
bicycle	0.296	0.315	**0.325**	0.000
mAP *as each source reports it*	0.472	0.472	0.616	0.130

Reading the results

1. **Agents of the most represented classes get most of the gradient.** Need for augmentation for less represented classes (future step).
2. **Query origin matters more than the fusion does.**
3. **Camera fusion brings small benefit with increased latency.** Camera fusion brings +40.3ms latency for only +0.025 EPA
4. **Reduction in number of queries** $\rightarrow$ **-9 ms latency**, but EPA drops 0.293 to 0.256 (savings cost)
5. **Reduction in number of sweeps beats ViP3D in every metric.**
6. **PointPillar LiDAR backbone is weak on detections.** Next step is to try Voxel Net (higher cost)

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Motivation

Vision-language-action models are a route to interpretable end-to-end driving. **Alpamayo-R1**[11] generates a **Chain-of-Causation reasoning trace**, and **flow-matching action expert** decodes a trajectory conditioned on it.

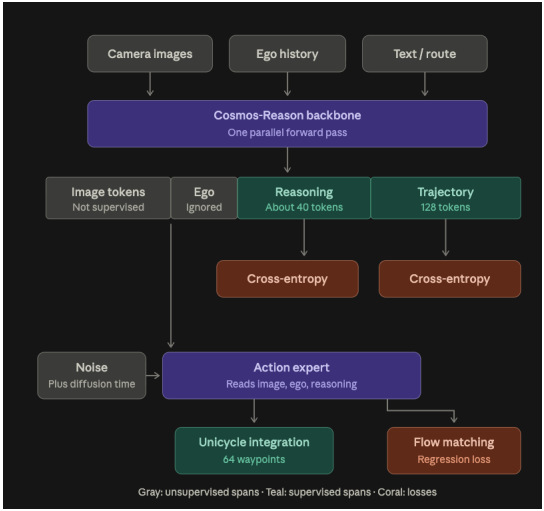
The reported gains concentrate exactly where imitation learning is weakest:

- 12 % improvement in planning accuracy on challenging cases
- 35 % reduction in close-encounter rate in closed-loop simulation.

Important structural observation

Trace is a functional intermediate representation between perception and planning. **End-to-end driving removed the hard interface** between perception and planning. AR1 restores an interface, but expresses it in natural language.

Alpamayo-R1 architecture



One backbone pass emits the reasoning and the trajectory tokens.

The grey spans carry no supervision.

The research gap

A trace says *"the black SUV from the left is going straight"*[11]. Nothing in the architecture requires that SUV to exist.

- The image tokens the reasoning conditions on are shaped **only by the language-modelling objective.**
- **Weak visual grounding** is listed among the limitations of supervised fine-tuning, and the mitigation is post-hoc, a reasoning model grading traces during RL (nothing constrains the reasoning at inference time)
- **No temporal preservation** - causal locality in the labelling protocol and a KV-cache rebuilt each frame.

Observation

Driving VLAs couple perception to reasoning through an unverifiable, memoryless interface, grounded in neither sensor evidence nor time.

Objectives and hypotheses

Objectives:

- Introduce an explicit, sensor-grounded, temporally propagated object representation into the VLA sequence
- evaluate whether it improves the quality of both the reasoning and the resulting trajectories

H1 **Grounding failure is measurable and non-negligible.** Traces from the released AR1 model refer to object categories absent from the scene at a rate significantly above that of the human-verified

H2 **An object-centric span reduces it.** Conditioning the reasoning on a span of detection-derived agent queries lowers that rate without degrading performance

H3 **Temporal propagation extends the benefit beyond the observation window.** Propagating queries across frames helps re-detection

Phase 1: measure the failure

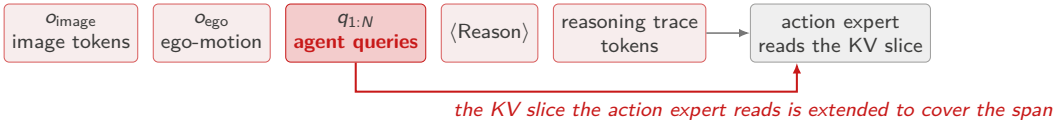
No training required. Run the released checkpoint over clips that also carry 3D annotations.

1. Parse each **model generated trace** for referenced categories and coarse positions.
2. Score a reference as unsupported when no GT annotated object of that category exists in that region at that time.
3. Apply the identical measurement to the **human-written traces**, compute the gaps between model and human traces

Note

It is cheap, it needs no new annotation, and it resolves whether the premise of the proposal holds.

Phase 2: the object span



Agent queries are projected into the backbone embedding space and inserted before the reasoning tokens. Causal attention then exposes every query to every reasoning token, so grounding reaches the language path and the control path at once.

The detection loss becomes the auxiliary objective AR1 asks for

Their future work[11] suggests depth or scene-flow auxiliaries to improve the visual backbone.

Controls, because the span might simply be ignored

Span masking Replace the queries with learned null embeddings at inference and measure the change in the generated traces. No change means the span is inert.

Span corruption Substitute queries taken from a different scene. A trace that adopts the foreign objects shows the span is genuinely read.

Phase 3: temporal propagation, and the grounding reward

depends on the time remaining

Temporal propagation

- Queries are propagated between frames with ego-motion compensation
- Each query keeps a FIFO memory bank of its own past states, attended over by temporal cross-attention [3]

Grounding reward

- Parse trace for object references
- Compare against the query set, score the match
- the RL stage gains a grounding reward

Example

Trace: *"yielding to the pedestrian on the right, lead vehicle stopped"*
Referenced: {pedestrian, vehicle} Query set: {car, car, truck}
Score: pedestrian unmatched, penalty applied.

Evaluation protocol and expected contributions

Evaluation.

- *Grounding*: unsupported-reference rate per category and scenario
- *Open-loop*: minADE_6 and minFDE over 6.4 s,
- *Closed-loop*: close-encounter rate, off-road rate and AlpaSim score, with an occlusion breakdown.
- *Cost*: added tokens, added parameters, end-to-end latency against the reported 99 ms budget.

Expected contributions.

1. The first quantitative characterisation of object-level hallucination in a released driving VLA, with a reusable protocol. This stands regardless of H2 and H3.
2. An architecture and training recipe for a sensor-grounded, temporally propagated object span, with ablations
3. A characterisation of the token-budget against grounding trade-off
4. A workshop or conference submission, with code released against the public checkpoints.

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How the stay builds on the thesis

The thesis produces exactly the representation AR1 lacks.

My thesis delivers

- A compact set of agent queries, tens rather than thousands
- Grounded in sensor data by a detection loss
- Carrying persistent identity across frames through a memory bank
- Initialised from the data rather

The stay requires

- A span short enough to insert into a token sequence
- A supervision signal that grounds it in the scene
- State that survives beyond the observation window
- Queries that reflect a particular scene

In one sentence

The thesis built the representation, which could be used to extend

Scientific questions to be addressed during the stay

1. How often does a released driving VLA refer to objects that are not there, measured against human-written traces on the same clips?
2. Does conditioning the reasoning on an explicit, detection-supervised object span reduce that rate, and can the span be shown to be used rather than ignored?
3. Does propagating the queries across frames extend the benefit to occlusion and reappearance, which causal locality and a per-frame KV-cache cannot reach?
4. What does the span cost in tokens, parameters and latency, and is that affordable on automotive-grade embedded hardware?
5. Can a rule-based grounding reward replace a judge model in the RL stage?

Fit with the Mobile Perception Systems Lab

- The lab's long-term goal is spatio-temporal reasoning that lets a mobile platform anticipate future events, identified as a bottleneck in crowded inner-city settings. **H3 is a concrete instance of that problem.**
- The test vehicle carries a 360-degree camera setup with LiDAR and radar, which enables the multimodal variant where LiDAR-derived queries ground an otherwise camera-only backbone. **My preliminary work is LiDAR-based and transfers directly.**
- Work in the lab on token-efficient vision transformers addresses precisely the cost an added span incurs, which the Phase 2 token-budget analysis measures.
- Radar perception suggests extending the multimodal variant to the conditions where a camera backbone is least reliable.
- Closed-loop evaluation would run in the TU/e digital twin alongside the public AlpaSim scenario sets.

Thank you

Technical discussion and questions

Backup: why EPA is the primary metric

EPA as defined by ViP3D[3]:

$$\mathrm{EPA} = \max\left(0,\ \frac{N_{\mathrm{hit}} - \alpha N_{FP}}{N_{\mathrm{GT}}}\right), \qquad \alpha = 0.5$$

- Predicted agents are matched to annotated agents by Hungarian assignment on centre distance, gated at 2 m, in the current frame.
- N_{hit} counts an agent as a hit only if its minFDE_K stays within 2 m at the horizon.
- minADE, minFDE and MR average only over matched agents, so a model that detects fewer objects can improve all three.
- N_{GT} counts the annotated agents that survive the whole horizon, and false positives are subtracted, so EPA penalises both failure modes at once.

Backup: the tracking interface

The model's inference behaviour is controlled by hard parameters

Parameter	Value
confirm a new track	0.40
keep a track alive	0.35
miss tolerance	5 frames
heatmap NMS kernel	3×3
active-track suppression	5×5 cells
queries K	300

Backup: batch size and query count sweep

Stage 1, LiDAR only, nuScenes validation. Learning rate was sqrt-scaled with the batch size.

Stage-1 variant	minADE ↓	EPA ↑	Note
300 queries, effective batch 8	0.956	0.293	best, the row in the results table
300 queries, batch 16	0.997	0.277	
300 queries, batch 32	1.114	0.236	
300 queries, batch 64	1.061	0.256	
100 queries, batch 8	1.022	0.256	
100 queries, batch 16	1.528	0.234	

Accuracy degrades close to monotonically with batch size even with the learning rate scaled, which is consistent with the usual small-batch-noise argument. I report it as an observation rather than an explanation, because the flat-minima claim is testable on the checkpoints I already have and I have not tested it yet.

Backup: configuration details

LiDAR branch

- Point cloud range ± 51.2 m, pillars of 0.2 m, 20 points per pillar.
- PillarFeatureNet $5 \to 64$, PointPillarsScatter to 512×512.
- SECOND with strides $[2, 2]$, SECONDFPN to $384 \times 256 \times 256$, so 0.4 m per BEV cell.
- 10 accumulated sweeps, 5 features per point.

Heatmap head

- CenterPoint nuScenes task grouping, so pretrained per-task weights still line up.
- Gaussian focal loss, radius from the box footprint.

Decoder

- Embedding 256, 8 heads, FFN 512, dropout 0.1.
- Layer 0 deformable BEV attention, 4 sampling points per query.
- Layer 1 SMCA over 4 FPN levels and 6 cameras, anisotropic σ from the projected box size.
- Reference points and sizes updated after each layer, sizes detached after layer 0.

Schedule

- Stage 1: 20 epochs, camera path frozen.
- Stage 2: 6 epochs, LiDAR path frozen, lr $3.5\mathrm{e}{-4}$.

Backup: questions for the lab

1. Which AR1 checkpoints and which parts of the Alpamayo training stack are available inside the lab, and is AlpaSim usable for closed-loop evaluation in February?
2. Do the campus recordings carry 3D annotations, or would Phase 1 run on the public clips and the campus data enter only in Phase 2 and 3?
3. What compute would a fine-tuning run have, and how is it scheduled?
4. Is the LiDAR and radar on the test vehicle time-synchronised and calibrated against the 360-degree camera rig to the accuracy that query-level projection needs?
5. Who else in the group works on the reasoning side, and would this sit alongside an existing effort or beside it?
6. What is a realistic publication target for a four-month result in your experience?

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